Review of: Bayesian Structural Time Series Analysis of Temperature Extremes

In this manuscript the authors develop a fully Bayesian framework for modeling non-stationarity in temperature extremes. They extend the existing literature on Dynamic Generalized Extreme Value (DGEV) models by addressing the well-known computational challenges that arise when fitting such models in slowly varying climate regimes, by adopting shrinkage priors and a non-centered parameterization. The observation layer retains the Generalized Extreme Value distribution for block maxima, while the latent state governing the location parameter evolves as a Gaussian random walk with level, slope and seasonal components. An approximate forward filtering backward sampling scheme based on local Laplace approximations is proposed as a computationally efficient alternative. The framework is applied to over a century of daily temperature records from Uccle, Belgium, and is used to produce forecasts of future temperature extremes. A structurally analogous Dynamic Linear Model is fitted in parallel to seasonal mean temperatures, allowing the authors to compare warming trends in the bulk and tail of the temperature distribution, though the two models are fitted independently rather than jointly.

Overall, I believe this is a well-engineered paper with real methodological contributions that will be useful to researchers working on non-stationary extremes. My main concerns are that the abstract overstates the connection between the bulk and tail models, and that it remains unclear whether the inferred temporal dynamics are genuinely learned from the data or largely reflect prior assumptions when evolution variances are near zero. I also believe the forecast evaluation could be strengthened by probing the tail of the predictive distribution more directly. I compiled these points in Major and Minor comments below and I believe that addressing them would strengthen the paper's contribution and make its claims more robust.

Major Comments

1. If I am reading the appendix correctly, the DLM for seasonal means and the DGEV for seasonal extremes are fitted as completely independent models – separate likelihoods, separate parameter vectors and separate FFBS passes. This is confirmed by the non-centered observation equation $\mu_t = c_t + H(Q)\tilde{x}_t$, where $H(Q)$ is a single row vector mapping to a scalar observation y_t. No information appears to flow between the two models during inference; even within the DGEV model, the linkage between bulk and tail is limited to shifts in the location parameter, with scale and shape held fixed. Considering this, I think describing the framework as providing a "unified yet separable parametric

view of bulk and tails" in the abstract risks overstating what the framework delivers. The models are structurally analogous, which is valuable, but I would not call that unified in a probabilistic sense. I think replacing "unified" with something like "structurally analogous" or "parallel" would more accurately reflect what the framework achieves and would make the genuine contribution, the shared decomposition into level, slope and seasonality, clearer rather than obscuring it behind a stronger claim.

2. The authors motivate the shrinkage prior as a data-driven alternative to tunable smoothing parameters in earlier work, which I find compelling in principle. However, when evolution variances q_k are near zero the likelihood for q_k may be relatively flat and uninformative. In that regime my concern is that the posterior over temporal dynamics may be driven primarily by the hyperparameters (a_λ, b_λ) rather than by the data. This matters because when $q_\alpha, q_\beta, q_\gamma$ are near zero, $H(Q)\tilde{x}_t \approx 0$ and the dynamic model effectively collapses toward the non-dynamic GEV_μ. If the shrinkage prior is driving this behavior more than the likelihood is, the credible intervals on exceedance probabilities and return levels may reflect prior assumptions rather than genuine data-driven uncertainty. I think comparing the posterior for each q_k against its prior and showing sensitivity of the inferred trajectories to the hyperparameter choice, would go a long way toward reassuring the reader that the dynamic model is genuinely learning from the data.
3. The approximate FFBS relies on the GEV log-likelihood being close to quadratic in μ_t, which I understand is a reasonable assumption for moderate observations. My concern is that this assumption is most likely to break down near the upper endpoint of the Weibull distribution, precisely where the paper draws its conclusions. For record-breaking observations like the 2019 Uccle value of 39.7°C, which sits close to the distributional endpoint given a shape parameter of around -0.29, the log-likelihood becomes strongly asymmetric rather than quadratic. Since this approximation is embedded within the forward filtering backward sampling scheme, any local deviation from quadraticity may propagate through the inferred latent trajectory. I think it would strengthen the paper considerably if the authors validated this approximation for the most extreme years in the record.
4. I appreciate that the choice of $\xi \sim U(-0.5, 0.5)$ is consistent with empirical guidance from climate and hydrology. However, with posterior estimates for maximum temperatures sitting around $-0.28\ to\ -0.31$, the lower bound of -0.5 is not far from the posterior mass. Since ξ directly governs the rate at which previously extreme events gain probability, I believe a brief sensitivity analysis showing how exceedance probability estimates change with different bounds would strengthen the paper's conclusions considerably. This would also help readers understand how robust the 2019 record probability estimate is to this modeling choice.

5. The cross-validation coverage is reported at the nominal 90% level, which checks whether observations fall within the central predictive interval. I think this is a reasonable starting point, but it does not directly assess the upper tail of the predictive distribution, the region beyond $\hat{q}_{\{0.95\}}$, which is where the paper's main claims about return levels and exceedance probabilities live. A model could in principle achieve perfect 90% coverage while badly misrepresenting the probability of events beyond that threshold. I believe reporting coverage at higher nominal levels such as 95% or 99% would give a much clearer picture of whether the model's tail predictions are reliable.
6. Looking at the annual extreme forecast panels in Figure F7, the predictive intervals appear nearly constant in width across the entire 20-year forecast horizon. In my understanding uncertainty about where the climate trend is heading over 20 years should produce intervals that widen progressively the further ahead you forecast. The nearly constant width suggests the model may be essentially extrapolating its estimated linear trend forward with fixed GEV observation noise on top, rather than genuinely propagating uncertainty about future warming. This is visually inconsistent with the paper's own finding that extreme events are becoming rapidly more likely; I would expect the upper bound of the predictive interval to widen more aggressively than the lower bound at long horizons. I think discussing this explicitly would strengthen the paper's treatment of forecast uncertainty.
7. If I understand correctly, the annual extreme forecasts are obtained by taking the maximum of 12 monthly predictive draws. It is not immediately clear to me that this is theoretically justified from an EVT perspective, since the maximum of already-extreme monthly observations is a different quantity from a standard block maximum of daily observations. I think a brief discussion of whether and why this aggregation is valid would add useful clarity for readers.

Minor comments

1. The manuscript would benefit from a careful proofread before final submission. There are several small typographical errors and formatting inconsistencies throughout. For example, "import" should read "important" in line 135, a period is missing in line 185, and there is an unnecessary indent in line 244.
2. I noticed that some variables are used before they are formally defined, which can make the manuscript harder to follow. For example, β appears in equation (5) before it is defined, and k is used in line 229 without a prior definition. I think making sure all variables are defined before or at their first appearance would improve the readability of the manuscript considerably.

3. The axis labels on several figures are quite small and difficult to read. I would encourage the authors to increase the font size of the axis labels throughout to improve readability, particularly for readers who may be viewing the manuscript in print.